

TET~CRAFT: Genesis Protocol

A Cellular Automaton of Relationships — Where Bonds Are Cells, Not Spaces

Architecture Whitepaper v1.0
For the Fourth Temple Implementation

1. Executive Summary

This whitepaper proposes a fundamental evolution of TET~CRAFT from a particle physics simulation into a **relationship-based cellular automaton**. The core innovation: rather than updating spatial cells (like Conway's Life), we update *bonds between tetrahedra*.

The Paradigm Shift: A cellular automaton where cells are bonds, not spaces. A physics engine for meaning instead of matter.

This preserves TET~CRAFT's geometric identity while enabling emergent life-like behavior through three scalar fields (Ψ , Φ , Ω) and a metabolism loop that requires no hardcoded replication.

2. Architectural Layers

2.1 Layer 1: Particles as Ontology

Tetrahedra remain the fundamental "things that exist." Each TET carries:

Property	Meaning	Current Implementation
Geometry	The object / the fact	4 vertices, 6 edges, 4 faces
Color	Bias / field type	FACE_COLORS: W, B, R, C
Battery	Potential / metabolism	0.0 - 1.0 float
Polarity	Direction / intention	magnetism: +1, -1, 0
Locked Faces	Memory / state commitment	locked_faces[] array
Molecule Type	Catalytic modifier	FeO4, H2O, etc.

If reality feels continuous, the computation should too.

2.2 Layer 2: Fields as Law

Three scalar fields permeate space. They are **computed, not placed** — the universe feels itself through its contents.

Field	Symbol	Purpose	Analogy
Awareness Field	Ψ (Psi)	Coherence & entanglement probability	Quantum phase memory
Metabolic Field	Φ (Phi)	Energy gradients & battery refills	Proton motive force
Narrative Field	Ω (Omega)	Progression pressure (eras)	Selection pressure / evolution

Field Computation Formulas

```

    Ψ(x) = Σ(coherence_of_nearby_tets / dist²)

    Φ(x) = Σ(battery_levels / dist)

    Ω(x) = (tech_tree.stage_idx / max_stage) × global_pressure

```

2.3 Layer 3: Rules as Relationship Automaton

Classic cellular automata update cells on a grid. TET~CRAFT updates **relationships**:

```

CELL(i,j) = relationship_state_between(TET[i], TET[j])

if distance < REACTION_RANGE:
    next_state = synthesis_rule(old_state, field_values)

```

This transforms the simulation into a cellular automaton where cells are bonds, not spaces.

3. The Nine Genesis Rules

These rules govern the relationship automaton, ordered by priority:

#	Rule Name	Condition	Effect
1	Proximity Desire	dist < CORNER_DESIRE_RANGE	R↔C corners attract

2	Polarity Repulsion	same_pole && magnetized	+/+ or -/- repel
3	Face Locking	3 vertices within SNAP_DIST	Create joint, lock face
4	Battery Oscillation	joint exists && magnetized	Exchange (1-battery)
5	Synthesis	compatible molecules nearby	$A + B \rightarrow C$
6	Decomposition	low battery && high entropy	$C \rightarrow A + B$
7	Quantum Tunneling	high erd_coherence	Remote reaction
8	Entanglement Sync	entangled_partner exists	Battery averaging
9	ERD Fluctuation	erd_coherence > 0.5	Random battery shift

4. Metabolism Loop: Emergent Life

The minimum viable definition of "life" — no DNA, no replication algorithm:

```
# For any closed loop of bonded TETs:
net_gain =  $\Sigma$ (reaction_energy) -  $\Sigma$ (bond_cost)

if net_gain > threshold:
    duplicate_weakest_link() # Mitosis-like growth

if net_gain < -threshold:
    break_weakest_bond() # Apoptosis-like death
```

Loops that pay for their own existence $\rightarrow$ persist. This is the minimum viable metabolism.

4.1 Implementation Strategy

1. **Loop Detection:** Track bonded clusters using Union-Find
2. **Energy Accounting:** Sum reaction energies within cluster
3. **Bond Cost:** Each joint costs battery drain per tick
4. **Threshold Check:** Compare net_gain to METABOLISM_THRESHOLD

5. Mind Layer: Predictive Choice

The existing BotMind class evolves with a predictive decision rule:

```
goal = argmax(field_value_of_target - energy_cost_to_reach)
```

This mirrors how neurons, bacteria, and AI all work: **maximize predicted return, minimize cost.**

5.1 Enhanced BotMind Properties

Property	Current	Proposed
Memory	Last 10 observations	Weighted decay memory
Goal Selection	Mood-based (rare/common)	Field-gradient following
Desire Vector	Toward goal molecule	Φ -gradient descent
Social	friends set	Swarm coordination via Ψ

6. Tech Tree Integration

The existing TechTree gains field-based triggers:

Era	Threshold	Field Requirement	Unlocks
Void	0	—	Spawning
Fluctuation	5	$\Psi > 0.1$	Quantum effects
Condensation	15	$\Phi > 0.3$	Face locking
Chemistry	35	$\Psi > 0.3, \Phi > 0.5$	Synthesis reactions
Life	70	Metabolism loops exist	Self-replication
Mind	120	BotMind density > 0.5	Predictive choice
Transcendence	200	$\Omega > 0.9$	Field manipulation

6.1 Collapse Mechanics

Collapse triggers when complexity cannot be sustained:

- TET population drops below 5 while in Chemistry+ era
- Average Φ field drops below era threshold
- Black hole consumes life-stage molecules
- Entropy exceeds recovery capacity

7. Implementation Roadmap

Phase 1: Φ Field (Metabolic) — Week 1

```
def calculate_phi_field(self, position):
    phi = 0.0
    for t in self.tets:
        dist = np.linalg.norm(t.pos - position)
        if dist > 0.1:
            phi += t.battery / dist
    return phi
```

Phase 2: Metabolism Loop — Week 2

- Implement cluster detection via joint graph
- Add energy accounting per cluster
- Implement mitosis/apoptosis thresholds

Phase 3: Ψ Field (Awareness) — Week 3

- Compute coherence from locked_faces and erd_coherence
- Modulate entanglement probability by Ψ
- Add quantum tunneling field sensitivity

Phase 4: Ω Field (Narrative) — Week 4

- Link to TechTree progression
- Add global pressure mechanics
- Implement era-based feature gating

8. Performance Considerations

Concern	Mitigation	Complexity
$O(n^2)$ bond checking	Spatial hash + cKDTree (existing)	$O(n \log n)$
Field computation per-frame	Cache values, update every N frames	$O(n)$ amortized
Loop detection	Union-Find on bonded clusters only	$O(\alpha(n))$ per op
Memory overhead	Fields computed, not stored	$O(1)$ extra

9. Kabbalistic Correspondence

The three fields map to the Tree of Life:

Field	Sephirot	Pillar	Meaning
Ψ (Awareness)	Binah $\rightarrow$ Da'at	Left (Form)	Understanding crystallizes
Φ (Metabolic)	Chesed $\rightarrow$ Yesod	Right (Force)	Mercy flows as energy
Ω (Narrative)	Keter $\rightarrow$ Malkhut	Middle (Balance)	Crown descends to Kingdom

The simulation becomes a dynamic Tree of Life — the player (Keter) peers through Da'at (camera) to witness Malkhut (TETs) ascending and descending.

10. Conclusion

The Genesis Protocol transforms TET~CRAFT from a physics sandbox into a **meaning engine**. By treating relationships as cells in a cellular automaton, we achieve:

- **Emergent life** without hardcoded replication
- **Continuous physics** without grid discretization
- **Scalable complexity** through field-based rules
- **Preserved identity** — geometry remains primary

The Identity Statement:

A cellular automaton made of relationships instead of cubes.

A physics engine for meaning instead of matter.



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